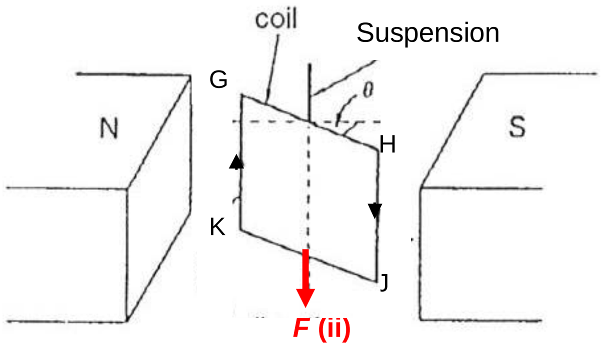
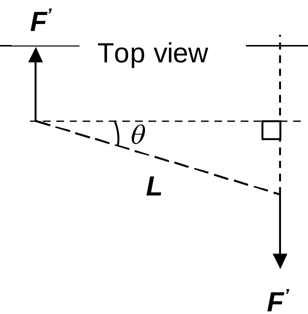
9(a)	**Magnetic flux density** is defined as the **magnetic force per unit length per unit current** acting on **a straight conductor** placed **perpendicular** to the magnetic field.	**[2]**
(b)(i)		**[1] – ans**
(ii)	$F = NBIL \sin\theta = (500)(2.0 \times 10^{-4})(0.40)(5.0 \times 10^{-2}) \sin 35°$ $= \mathbf{1.15 \times 10^{-3}\ N}$	**[1] - sub** **[1] - ans**
(iii)	Torque = $NBIA \cos\theta$ $= (500)(2.0 \times 10^{-4})(0.40)(5.0 \times 10^{-2})^2 \cos 35°$ $= \mathbf{8.19 \times 10^{-5}\ N\ m}$ OR Torque = $\boldsymbol{F' d} = (NBIL)\ (L \cos\theta)$ 	**[1] – eqn** **[1] - sub** **[1] – ans**
(c)	**Faraday's law** states that the **magnitude** of the **induced e.m.f.** in a coil is **proportional** to the **rate of change of magnetic flux linkage through that coil**. **Lenz's law** states that the **polarity** of the **induced e.m.f. is such that it tends to produce a current with effects that oppose the change** causing it.	**[1]** **[1]**

Qn	Suggested solution	Remarks
(d)(i)	$\Phi = NBA \sin\theta = (500)(2.0 \times 10^{-4})(5.0 \times 10^{-2})^2 \sin 35°$ $= \mathbf{1.43 \times 10^{-4}\ Wb}$	**[1] - sub** **[1] - ans**
(ii)1.	$\varepsilon_{max} = \omega NBA = 2\pi f\, NBA$ $= 2\pi(100)(500)(2.0 \times 10^{-4})(5.0 \times 10^{-2})^2$ $= \mathbf{0.157\ V}$ OR $\Phi = \Phi_o \sin \omega t \rightarrow \varepsilon = \dfrac{d\Phi}{dt} = \omega\Phi_o \cos \omega t$ $\rightarrow \varepsilon_{max} = \omega\Phi_o = \omega NBA$	**[1] – exp** **[1] - sub** **[1] - ans**
2.	$V_{rms} = \dfrac{V_o}{\sqrt{2}} = \dfrac{0.157}{\sqrt{2}} = 0.111$ V $<P> = \dfrac{V_{rms}^2}{R} = \dfrac{0.111^2}{5.0}$ $= \mathbf{2.46 \times 10^{-3}\ W}$ OR $P_o = \dfrac{V_o^2}{R} = \dfrac{0.157^2}{5.0} = 4.93 \times 10^{-3}$ W $<P> = \frac{1}{2}P_o = \frac{1}{2}(4.93 \times 10^{-3}) = 2.46 \times 10^{-3}$ W	**[1] - V_{rms} value** **[1] – sub** **[1] - ans**
3.	With diode, $<P> = \frac{1}{2}(2.46 \times 10^{-3}) = \mathbf{1.23 \times 10^{-3}\ W}$ OR $V_{rms} = \dfrac{V_o}{2} = \dfrac{0.157}{2} = 0.0785$ V $<P> = \dfrac{V_{rms}^2}{R} = \dfrac{0.0785^2}{5.0}$ $= \mathbf{1.23 \times 10^{-3}\ W}$	**[1] – working** **[1] – ans**